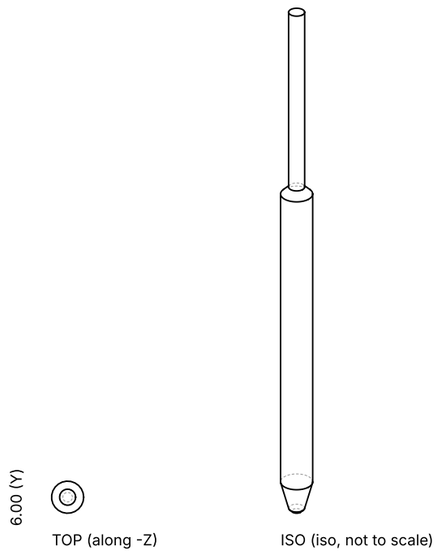

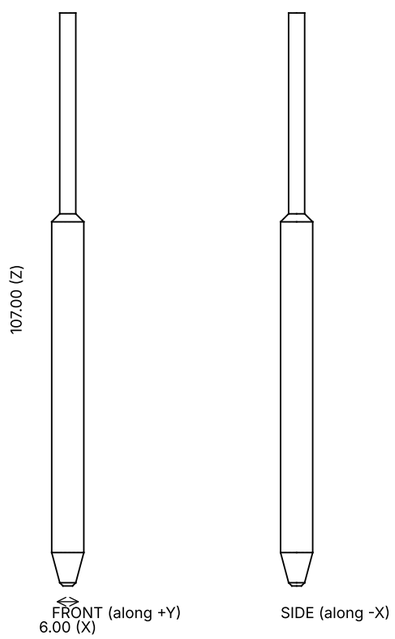

ENG-TC-05   Pintle needle (15 deg metering cone, 6 mm body, 3 mm stem)
Material: 17-4PH (1.4542) condition H1150, pins   Process: CNC lathe, stem ground h7 Ra 0.2
Mass: 16.6 g (CAD volume x density)   Units: mm   Frame: modelling frame (+Z forward)   Bounding box: 6.00 x 6.00 x 107.00
Datum: seat contact line A   Finish: passivate (ASTM A967)   Heat treatment: 17-4PH H1150
Fits / features: stem 3 h7 (0/-0.010) Ra 0.2; M3-6g top thread, 6 mm; body 6 f7; cone 15 deg +/-5' lapped to TC-03
Notes: closed position shown; stroke 4.2 mm away from the chamber opens (crank stop 113.09 deg); cone half-angle 15 deg (combustion.json); v3.4 stem 3 h7 through the PTFE spring-energised seal (TC-04), M3 x 0.5 top thread 4.5 m
ARGOS engine CAD v3.5.1 - DESIGN-MODEL DRAWING generated from the solid; dimensions from the model; general tolerances ISO 2768-m unless a fit is listed; NOT released for fabrication (Mechanical, 2026-10-13)
Oxygen service: clean to ASTM G93, Krytox only